MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 5696

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode, xenon gas

Mounting position: Any

Weight: 0.3-ounce (8.5 grams) nominal

Base: E7-1 (EIA)

Envelope: T5-1/2 (EIA)

Base connections:

Pin No.:	1	2	3	4	5	6	7
Element:	g1	k	h	h	g2	a	g2

ABSOLUTE RATINGS:

				Nonconduction		Conduction		
Parameter:	Ef	epx	epy	ec2	ec1	Ec2	Ec1	Ic1
Unit:	V	v	v	v	v	Vdc	Vdc	mAdc
Maximum:	6.9	500	500	-50	-100	-10	-10	5
Minimum:	5.8	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	---	---	0	---	---	---	(Note 1)

ABSOLUTE RATINGS:

Parameter:	Ic2	ik	Ik	Ehk	TA	tk	Rg1
Unit:	mAdc	ma	mAdc	V (Note 2)	°C	sec	Meg
Maximum:	5	100	25	-100	+90	---	10
Minimum:	---	---	---	---	-75	10	---
TEST CONDITIONS:	(Note 1)	---	---	25	---	---	---

GENERAL:

Qualification - Required

(D) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3201	Critical grid voltage for conduction (1)	3,4	Epp = 350 Vac; Rgl = 0.1 Meg; Rp = 5,000 ohms	0.65	II	Eccl	-2.3	-3.9	Vdc
3201	Critical grid voltage for conduction (2)	3,4	Epp = 350 Vac; Rgl = 10 Meg; Rp = 5,000 ohms	0.65	II	Eccl	0	-4.8	Vdc
1231	Pulse emission	-	e = 180 ±9v; prr = 100 ±5; Ra = 8.6 ohms; Rp = 40 ohms; Zm = 7.5; t = 3 seconds (max); calibrating resistor = 20 ohms	0.65	II	etd	---	75	v
(D)3201	Critical anode voltage for conduction (1)	3	Vary Ebb; Ec1 = 0; Rgl = 0.1 Meg; Rp = 3,000 ohms	0.65	II	Ebb	---	32	Vdc
(D)3201	Critical anode voltage for conduction (2)	3,5	Vary Ebb; Ef = 0; Ec1 = -100 Vdc; Rgl = 0; Rp = 10,000 ohms	0.65	II	Ebb	500	---	Vdc
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	-	No voltages applied	---	---	---	---	---	---
3241	Heater current			---	---	If	138	162	mA
1336	Heater-cathode leakage	-	Ehk = -100 Vdc Ehk = +25 Vdc	---	---	Ihk Ihk	0 0	20 20	µA µA
3201	Critical grid voltage for conduction (3)	3,4,6	Ef = 7.0 V; Epp = 350 Vac; Rgl = 10 Meg; Rp = 5,000 ohms	---	---	Eccl	0	-5.6	Vdc
	Quality conformance inspection, part 3								
---	Life test	7	Group B; Ib = 25 mAdc; ib = 150 ma; Rgl = 5,000 ohms; Epp = 350 Vac; t = 500 hours	---	---	---	---	---	---
---	Life-test end point:								
1231	Pulse emission	-		---	---	etd	---	100	v

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NOTES:

1. Over a 30-second maximum interval.
2. Cathode is reference element.
3. Connect pins 5 and 7 to pin 2.
4. Use minature steatite socket with grounded shield base. Shield the anode power supply. Use short shielded anode and grid leads. Anode and grid resistors should be noninductive. Connect the grid resistor directly at the socket.
5. No voltages shall be applied to the tube for a period of 20 minutes minimum preceding the test.
6. Preheat for a period of 15 minutes under test conditions or under conditions: Ef = 7.0; Epp = vary; Ec1 = Ec2 = 0; Rg = 10 Meg; Ib = 100 mAdc. Five seconds maximum between preheat and test.
7. Phase of grid voltage adjusted to provide start to conduction at peak of applied anode voltage.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2922)

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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NAVY DEPARTMENT
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DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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